

Dataset Analysis for Decision-Making

March 20, 2025

 **sicor_municipal_co... .csv**
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address your query, I will first load the `sicor_municipal_contracts.csv` file to understand its structure and key characteristics. Once I have a clear understanding of the dataset, I can proceed to answer your specific analytical question.

Below is the step-by-step process:

Step 1: Load and Inspect the Dataset

I will load the dataset using Python's `pandas` library and inspect its structure, including column names, data types, and a few sample rows. This will help identify the key characteristics of the dataset.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
```

Step 2: Identify Key Characteristics

From the output of the above code, I will identify:

1. **Columns** : The names and descriptions of the columns.
2. **Data Types** : Whether the columns contain numerical, categorical, or date/time data.
3. **Missing Values** : Any missing or null values in the dataset.
4. **Unique Values** : The number of unique entries in categorical columns.
5. **Summary Statistics** : Basic statistics for numerical columns (e.g., mean, min, max).

Step 3: Address Your Analytical Question

Once you provide a specific analytical question, I will write Python code to extract and process the relevant data from the dataset to answer it. The deliverables will include:

1. A clear, concise answer to the question.
2. The Python code used to derive the answer.

Next Steps

Please provide the specific analytical question you would like me to address using the dataset. For example:

- What is the total value of contracts awarded by each municipality?
- Which contractor received the highest number of contracts?



- Are there any trends in contract values over time?

Let me know your question, and I will proceed with the analysis.

How can the combination of characteristics of each activity increase the total contract value? Define three options to address this issue and select the best one.



To address your question, we need to analyze the dataset and explore how the characteristics of each activity (e.g., contractor, municipality, activity type, duration, etc.) influence the total contract value. The goal is to identify patterns or combinations of characteristics that could potentially increase the total contract value.

Step 1: Understand the Dataset

From the dataset, we will focus on columns that describe the characteristics of activities and the total contract value. These might include:

- **Contractor** : The entity responsible for executing the activity.
- **Municipality** : The location where the activity is performed.
- **Activity Type** : The nature of the activity (e.g., construction, maintenance).
- **Duration** : The time allocated for the activity.
- **Budget/Value** : The monetary value associated with the activity.

We will analyze these characteristics to determine their impact on the total contract value.

Step 2: Define Three Options to Increase Total Contract Value

Based on the dataset, here are three potential options to increase the total contract value:

Option 1: Focus on High-Value Contractors

Identify contractors who consistently secure high-value contracts and allocate more activities to them. This assumes that experienced or specialized contractors can deliver higher-value projects.

Option 2: Target High-Income Municipalities

Allocate more activities to municipalities with a history of awarding high-value contracts. This assumes that wealthier or more developed municipalities have larger budgets.

Option 3: Prioritize High-Impact Activity Types

Focus on activity types that historically yield higher contract values. For example, large-scale infrastructure projects might have higher budgets compared to routine maintenance.

Step 3: Analyze the Dataset to Select the Best Option

We will analyze the dataset to evaluate which option has the greatest potential to increase the total contract value.

```
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5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
14
15 # Ensure relevant columns exist in the dataset
16 required_columns = ['contractor', 'municipality', 'activity_type', 'contract_value']
17 if not all(col in data.columns for col in required_columns):
18     raise ValueError(f"Missing required columns: {set(required_columns) - set(data.columns)}")
19
20 # Convert contract_value to numeric if it's not already
21 data['contract_value'] = pd.to_numeric(data['contract_value'], errors='coerce')
22
```



```

23 # Option 1: Focus on High-Value Contractors
24 contractor_summary = data.groupby('contractor')['contract_value'].sum().sort_values(ascending=False)
25 top_contractors = contractor_summary.head(5) # Top 5 contractors by total contract value
26
27 # Option 2: Target High-Income Municipalities
28 municipality_summary = data.groupby('municipality')['contract_value'].sum().sort_values(ascending=False)
29 top_municipalities = municipality_summary.head(5) # Top 5 municipalities by total contract value
30
31 # Option 3: Prioritize High-Impact Activity Types
32 activity_summary = data.groupby('activity_type')['contract_value'].sum().sort_values(ascending=False)
33 top_activities = activity_summary.head(5) # Top 5 activity types by total contract value
34
35 # Display results
36 print("\nTop Contractors by Total Contract Value:")
37 print(top_contractors)
38
39 print("\nTop Municipalities by Total Contract Value:")
40 print(top_municipalities)
41
42 print("\nTop Activity Types by Total Contract Value:")
43 print(top_activities)
44
45 # Determine the best option based on the highest total contract value
46 best_option = None
47 if top_contractors.sum() >= top_municipalities.sum() and top_contractors.sum() >= top_activities.sum():
48     best_option = "Focus on High-Value Contractors"
49 elif top_municipalities.sum() >= top_contractors.sum() and top_municipalities.sum() >= top_activities.sum():
50     best_option = "Target High-Income Municipalities"
51 else:
52     best_option = "Prioritize High-Impact Activity Types"
53
54 print(f"\nThe best option to increase total contract value is: {best_option}")

```

Step 4: Interpretation of Results

- Option 1 (High-Value Contractors)** : If specific contractors consistently secure high-value contracts, allocating more resources to them could maximize the total contract value.
- Option 2 (High-Income Municipalities)** : If certain municipalities award significantly higher-value contracts, focusing efforts on these areas could be beneficial.
- Option 3 (High-Impact Activity Types)** : If specific activity types (e.g., infrastructure projects) yield higher contract values, prioritizing these activities could increase the total value.

The Python code calculates the total contract value for each option and selects the one with the highest potential.

Deliverables

- Answer** : The best option to increase the total contract value is determined by the analysis. For example:
 - If contractors dominate, the answer is "Focus on High-Value Contractors."
 - If municipalities dominate, the answer is "Target High-Income Municipalities."
 - If activity types dominate, the answer is "Prioritize High-Impact Activity Types."
- Python Code** : The provided code extracts and processes the data to derive the answer.

Please run the code and share the results if you'd like me to interpret them further!

